

MRI RF COILS – HOW THEY WORK & WHAT ENGINEERS NEED TO KNOW

RF coils are the interface between the patient and the MRI system. They transmit RF energy to excite spins (B_1^+) and/or receive the tiny MR signal (B_1^-).

GOOD COILS → Strong Signal (High SNR) • Uniformity • Proper Coverage • Patient Safety

1. COIL ROLES IN MRI

TRANSMIT (TX)

Create the B_1 field that excites the spins.



- Deliver RF power
- Set flip angle
- Usually lower frequency (body coil)
- High power handling

RECEIVE (RX)

Detect tiny MR signals from the patient.



- Extremely small voltages (μV)
- High sensitivity (preamp close)
- Low noise is critical
- Determines SNR

Many coils are TX/RX – they do both (not at the same time).

2. COIL TYPES OVERVIEW

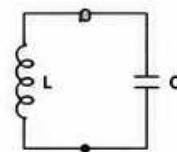
Coil Type	Transmit (TX)	Receive (RX)	Typical Use	Examples
Body Coil	✓	✓	Whole-body transmit & receive	GE HD Body Coil (Integrated)
Transmit-Only Coil (TX)	✓	–	High power specialty transmit	Birdcage TX Coil
Receive-Only Coil (RX)	–	✓	Maximum SNR for local imaging	Spine Array (RX Only)
TX/RX Coil (Single Element)	✓	✓	Small coverage (single channel)	Knee Coil
Array Coil (Multi-Element)	–	✓	High SNR + parallel imaging (multi-channel)	Head 32-Ch Array

Most modern imaging uses multi-channel receive arrays with body coil transmit.

3. HOW A COIL WORKS (PRINCIPLES)

RESONANCE

Each coil is tuned to the Larmor frequency (f_0).



$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

ELECTROMAGNETIC INDUCTION

• Transmit: RF current → B_1 field → excites spins.



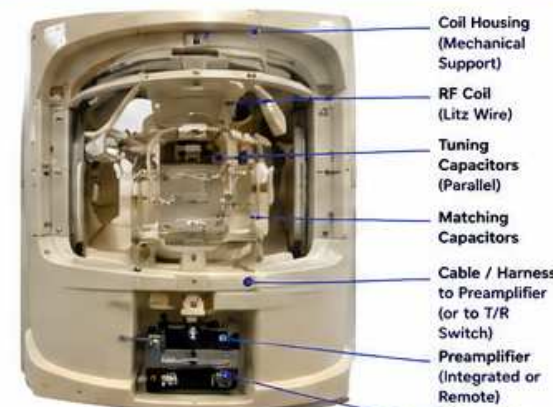
• Receive: Precessing spins → induce voltage in coil.



KEY POINT

Coils must be tuned (resonant) and matched (impedance) for maximum power transfer (TX) and maximum signal (RX).

4. COIL COMPONENTS (TYPICAL)

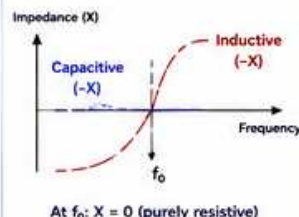


All components affect performance. A change in any part can impact tuning, matching, decoupling and SNR.

5. TUNING & MATCHING EXPLAINED

TUNING (Resonance)

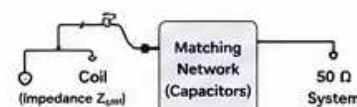
Adjust capacitors so the coil resonates at f_0 (Larmor frequency).



At f_0 : $X = 0$ (purely resistive)

MATCHING (Impedance)

Adjust matching capacitors so the coil impedance (Z) matches the system (typically 50 Ω).



Maximum power transfer (TX)
Maximum signal transfer (RX)

GOOD TUNING & MATCHING RESULTS IN:

- ✓ Maximum transmit efficiency
- ✓ Maximum received signal (SNR)
- ✓ Proper flip angle
- ✓ Reduced heating
- ✓ Best image quality

6. COIL LOADING (PATIENT EFFECT)

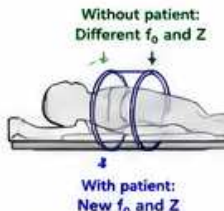
The patient's body changes the coil's impedance.

Effects:

- Shifts resonance frequency (f_0)
- Changes impedance magnitude (Z)
- Reduces Q (quality factor)
- Affects tuning and matching

Implication:

Always tune and match with the patient (or a proper phantom) in place whenever possible.

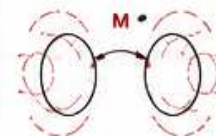


7. COIL COUPLING & DECOUPLING

In array coils, elements can couple to each other.

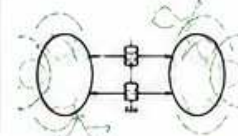
COUPLING (Bad)

- Elements share signal
- Noise increases
- Parallel imaging suffers



DECOUPLING (Good)

- Elements are isolated
- Noise reduced
- SNR and parallel imaging improved



8. COIL METRICS THAT MATTER

SNR (Signal-to-Noise Ratio)
Higher is better.

Q (Quality Factor)
Higher Q → better efficiency and SNR (too high Q can be narrow bandwidth).

Bandwidth
Should be wide enough for the pulse.

Noise Figure (NF)
Lower NF → better SNR.

g-Factor (Array Coils)
Lower g → better parallel imaging performance.

B_1 Field Uniformity (TX)
More uniform → consistent flip angle.



9. COMMON COIL CONFIGURATIONS (EXAMPLES)

BODY COIL (TX/RX)

Large volume coverage
Lower SNR (far from patient)



Used for whole body imaging and transmit.

SPINE ARRAY (16-CH RX)

Multiple elements along spine
High SNR & parallel imaging



Best near the elements. SNR falls with distance.

HEAD ARRAY (32-CH RX)

High channel count
Excellent SNR



Close fit + many elements = high SNR & detail.

KNEE COIL (TX/RX)

Small FOV, high fill factor
High SNR



Small coils give highest SNR for local imaging.

10. COMMON COIL PROBLEMS & WHAT THEY LOOK LIKE

Problem	Symptoms / Image Appearance	Cause (Examples)	Check / Action
Detuning Failure in TX	Severe RF noise / "zipper" / banding across image	PIN diode open, cable issue, detuning trap failed	Check detuning diodes, traps, cable continuity
Poor Tuning	Low SNR, shading, frequency error messages	Tuning caps shifted, loose connections	Retune with patient/phantom
Poor Matching	High reflected power, low flip angle, low SNR	Matching caps off, cable fault, connector issue	Check match, cables, connectors
Element Failure (Array Coil)	Signal void / dropout in a specific area	Open element, preamp failure, cable break	Run coil element check, inspect cables
High Noise Figure	Overall low SNR	Preamp failure, noise source	Check preamp, power, cabling, grounding

11. COIL CARE & BEST PRACTICES

- Keep connectors clean and dry.
- Use coil covers to keep fluids out.
- Never bend or twist cables.
- Do not drop or strike coils.
- Retune and match after any repair.
- Always test with phantom and patient.



ENGINEER NOTES

- Always follow ESD precautions when handling coil components.
- Document original tuning/matching settings before adjustments.
- Use approved test equipment (network analyzer, power meter, etc.).

ABBREVIATIONS

RF = Radio Frequency TX = Transmit RX = Receive T/R = Transmit / Receive
L = Inductance C = Capacitance f_0 = Larmor Frequency SNR = Signal-to-Noise Ratio
Q = Quality Factor Z = Impedance NF = Noise Figure FOV = Field of View



THE BOTTOM LINE

Well-designed, properly tuned, and healthy RF coils are essential for:
High SNR • Uniform images • Accurate flip angles • Patient safety



GE HealthCare
Imagination at work